


SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

# Lab 6 – Gravity Gradient perturbation

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# Second Tutoring session

Friday 24 October

Room L.11

14:15 – 16:15

Due to Wednesday 22nd - Write questions to me: annachiara.ippolito@polimi.it

These hours are meant to fill the gap on the fundamentals of MATLAB and Simulink.
**Partecipate if you have questions: it will not be a lecture but an interactive session!**

www.tutorapp.polimi.it -> try doing a first access!

The day of the tutoring you must register **within the first half an hour**.

My person code: 10661425

# Task 1: Compute the relative attitude between the S/C body frame and a moving reference frame

**Principal moments of Inertia**

$I_x = \{0.040, 0.060\}\ \mathrm{kg\,m^2}$
$I_y = \{0.060, 0.080\}\ \mathrm{kg\,m^2}$
$I_z = \{0.080, 0.040\}\ \mathrm{kg\,m^2}$

**Initial conditions**

$\omega_x(0) = \{0, 10^{-6}\}\ \mathrm{rad/s}$
$\omega_y(0) = \{0, 10^{-6}\}\ \mathrm{rad/s}$
$\omega_z(0) = n\ \mathrm{rad/s}$
$\mathrm{A}_{B|N}(0) = \mathrm{A}_{\mathrm{L|N}}(0)$

**Governing Equations**

*Dynamics*

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\,\omega_y\omega_z \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\,\omega_x\omega_z \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\,\omega_y\omega_x \end{cases}$$

*Kinematics*

$$\frac{\mathrm{dA}_{B|N}(t)}{\mathrm{d}t} = -[\omega]^{\wedge}\mathrm{A}(t)$$

*Reference Attitude*

$$\mathrm{A}_{L|N} = \begin{bmatrix} \cos(nt) & \sin(nt) & 0 \\ -\sin(nt) & \cos(nt) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{\omega}_{L|N} = \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix} \quad \text{with} \quad n^2 = \frac{GM_t}{R^3}$$

You have a spacecraft orbiting Earth at a certain altitude $H$. You want the spacecraft to follow the reference moving attitude given by $\mathrm{A}_{\mathrm{L|N}}$ and $\boldsymbol{\omega}_{\mathrm{L|N}}$, but currently you have no controls on your satellite.

**Tasks**

- Compute the attitude error between the S/C orientation and the reference orientation at each timestep.

| | |
|---|---|
| Attitude error | $\mathrm{A}_{B\|L} = \mathrm{A}_{B\|N}\mathrm{A}_{L\|N}^{\mathrm{T}}$ |
| Ang. vel. error | $\boldsymbol{\omega}_{B\|L} = \boldsymbol{\omega}_B - \mathrm{A}_{B\|L}\boldsymbol{\omega}_L$ |

# Task 2: Repeat including the Gravity Gradient perturbation

## Governing Equations

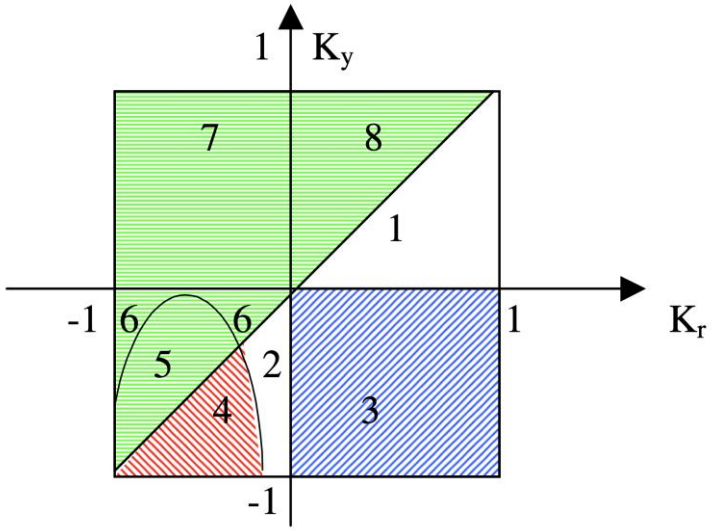
### *Dynamics*

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\,\omega_y\omega_z - \dfrac{3GM_t}{R^3}\dfrac{I_y - I_z}{I_x}\,c_2c_3 \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\,\omega_x\omega_z - \dfrac{3GM_t}{R^3}\dfrac{I_z - I_x}{I_y}\,c_1c_3 \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\,\omega_y\omega_x - \dfrac{3GM_t}{R^3}\dfrac{I_x - I_y}{I_z}\,c_2c_1 \end{cases}$$

with $\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \mathrm{A}_{B|L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (try to use vectorial form!)

## Tasks

- Simulate the spacecraft dynamics and kinematics under the effect of the gravity gradient perturbation.
- Compute the attitude error with respect to the reference attitude
- Validate some points in the stability region diagram




$$K_x = \frac{I_z - I_y}{I_x}$$
$$K_y = \frac{I_z - I_x}{I_y}$$
$$K_z = \frac{I_y - I_x}{I_z}$$